Mind World

A branching, scroll-scrubbed portfolio. Design & build specification.

`arpit2412.github.io` · worktree `~/Desktop/Resume/site-world` · branch `world/v2`
Drafted 2026-07-10 · Status: **blocked on assets** (photos + Higgsfield credits)

1. The concept

The page opens on Arpit, full body, standing. The visitor scrolls; the camera pushes forward into his head. Inside is a **hub** — a chamber with doors. Each door is a domain of the work: Research, Engineering, Film & VFX, Writing, Honors. Choosing one flies the camera into that room, where the work lives. Scrolling back up surfaces you through the hub, where you can choose again.

It is not a linear tour. It is a space you re-enter. Every room is reachable from every other room, because every room returns you to the hub first, and the hub is one move from anywhere.

> **Branching costs less than linear, not more.** Scrubbing is bidirectional: scrolling backwards through a clip plays it in reverse. The *hub* → *Engineering* flight, reversed, *is* the *Engineering* → *hub* flight. Every return journey is free. Five rooms therefore need five edges, not ten. The branching graph costs **74 credits**; the linear eight-chapter version cost **91**.

2. Structure

```
            [ PORTRAIT ]  full body, locked off
                  |  scroll → camera pushes into the head
                  v
             [  HUB  ]  a chamber, five doors, ambient
              /  |  |  \  \
             /   |  |   \  \        each edge = ONE generated clip
            v    v  v    v  v       reverse-scrub gives the return free
      RESEARCH  |  |   WRITING  HONORS
              ENGINEERING
                 FILM & VFX

   surfacing from any room = that room's edge, played backwards
   → any room is two moves from any other room, via the hub
```

Room contents (all existing content preserved)

ROOM	ANCHOR	HOLDS
Research	`#research`	Six research areas, selected publications
Engineering	`#work`	Project cards, the Lifecycle block (pretrain → unlearn)
Film & VFX	`#films`	Nine films, TikTok / AI Parameters content
Writing	`#writing`	Blog posts, news
Honors	`#honors`	Patents (US + UK), grant, awards
Journey	`#journey`	Career timeline — surfaces in the hub itself

3. Cost

All figures below were measured with live `get_cost` preflights against the Higgsfield MCP, not estimated.

ITEM	QTY	UNIT	CREDITS
Stills — `nano_banana_pro` , 2K, 16:9	7	2	14
Dive clips — `cinematic_studio_video_v2` , 5s, silent	7	5	35
Hub → room edges	5	5	25
Room → hub edges *(reverse-scrub)*	5	0	0
Base total			**74**
With realistic 2× rerolls			148

Model choice matters more than plan choice

VIDEO MODEL	PER 5S CLIP	BASE BUILD	FITS STARTER (270CR)?
`cinematic_studio_video_v2`	5.0	74	Yes, with 3× headroom
`seedance_2_0` fast, 720p	17.5	224	Base only, zero rerolls
`seedance_2_0` std	22.5	284	No — overshoots

> **Starter is a gamble on model access, not on credits.** Starter's own feature list says “access to *selected* models” and names only Seedance 2.0 Fast / Mini. If `cinematic_studio_video_v2` is not included, the build cannot complete on 270 credits. **First spend must be one 5-credit test clip on that exact model.** If it is refused, upgrade to Plus.

> **MCP and web prices disagree.** The MCP checkout offers Plus at **$49/mo (1,000 cr)** or **$29/mo annual**; higgsfield.ai advertises $59/mo for 1,200. Also: the MCP's own compliance note states *“unlimited models and free generations are accessible only via higgsfield.ai and are not accessible on MCP/CLI.”* All “7 unlimited models” bullets are worth nothing for this build — it runs through the MCP. Compare tiers on raw credits and model unlocks only.

4. Blockers

B1 — No credits. `list_workspaces` returns one workspace: `plan_type: "free"`, `credits: 0`. The MCP is connected and authenticated; it simply has no balance. Nothing generates until this is resolved. *Owner: Arpit.*

B2 — No photographs. The repo contains one image, `/arpit.png`. The concept requires Arpit's likeness to survive seven generated scenes. Two routes:

- `image_to_3d` → a real GLB mesh, orbitable, but needs `three.js` and reproduces only what is in the source photo.
- **Soul training** (5–20 photos, ~10 min) → a consistent face across every shot. Recommended — “inside my mind” fails if the face drifts between rooms.

Needed: full-body photos, several angles, good light. *Owner: Arpit.*

B3 — Film footage is not ours to publish. Furiosa, Deadpool, Mickey 17, Mortal Kombat II and the rest are owned by Warner Bros., Disney, Netflix et al. VFX work at Rising Sun Pictures is normally under NDA. Playing film clips or GIFs on a public portfolio is a live exposure, not a theoretical one.

The existing site already handles this correctly — `data/films.ts` uses Wikipedia poster art plus license-free Unsplash “mood” banners, and says so in a comment. **Keep that.** Arpit's own TikTok / AI Parameters content is his and can play freely. For the films: poster art plus a line on what shipped, or generated plates that evoke the work without being the work. *Owner: Arpit — decision required.*

5. Engineering

What survives

`scrubEngine.ts` — forked from scroll-world (MIT, oso95/scroll-world). It is segment-based and does not care what the chapters are, so the graph rewrite does not touch it. Two bugs were fixed against upstream:

- **Frame-rate-dependent smoothing.** Upstream runs `cur += (target - cur) * 0.18` once per rAF, which converges twice as fast at 120 Hz as at 60 Hz. Now solved for real frame delta.
- **Layout thrash.** Upstream measured element offsets inside the scroll handler. Offsets are now cached and recomputed on resize / ResizeObserver.

Kept as a lerp, never a spring: spring overshoot on a media seek scrubs past the target frame and visibly rewinds.

What must be rebuilt

- The chapter *chain* becomes a small *graph* with a router. Hub is a choice state; each room is its own scroll track; deep links (`#engineering`) drop straight in. This is real work, not a config change.
- Reverse-scrub as the return journey needs explicit handling at the hub boundary.

> **Opacity conflict.** On `origin/master`, `Research`, `Timeline` and `Writing` are `bg-surface` — fully opaque — and every card is `bg-card`. A fixed video stage behind them shows through in only five of nine sections. That alternating opacity *is* the current design's banding rhythm. Making the video visible everywhere is a deliberate redesign, not a bug fix.

> **Film grain does not compress.** Synthetic full-frame temporal noise at the pipeline's specified `crf 20` produced **273 MB** across 26 short placeholder clips. Generated footage behaves the same. Mitigate with low-amplitude grain in prompts, AV1/HEVC (grain synthesis) with an h264 fallback, or a CDN.

Hosting

`arpit2412.github.io` is a GitHub Pages hostname and **cannot** be pointed at Vercel. Moving hosts means a new domain. Nothing in the build assumes a host; the engine fetches clips as same-origin Blobs, so an absolute CDN URL needs CORS on the bucket.

6. Order of work

1. Arpit: supply photos; decide the films question; buy credits.
2. **2 credits** — one still of Arpit. He approves the face before anything else is spent.
3. **5 credits** — one test clip on `cinematic_studio_video_v2`. Proves model access *and* motion quality.
4. Remaining ~67 credits — the other six stills, then the dives, then the five hub edges.
5. Wire the graph router; rebuild the chapter map against the real components.
6. Encode desktop + mobile variants; decide repo vs CDN on measured bytes.

Stills are rerolled at 2 credits; clips at 5. Approve every still before generating any motion — it is the cheapest place to be wrong.

7. Record of corrections

Three mistakes were made and corrected. They are logged so they are not repeated.

- **Built against a stale checkout.** The local tree was 5 commits behind `origin/master`; the live site is a ground-up redesign. `git log` was run; `git fetch` was not. Components that were wired (`Publications.tsx`, `FilmsMarquee.tsx`) no longer exist upstream. The stale tree also carries a wrong grant figure — A$2.1M, corrected upstream to **A$1.2M**.
- **Designed without watching the reference.** The scroll-world demo is a reel of *three* sites — an isometric diorama *and* two photoreal ones (luxury real estate, automotive). The claim that “photoreal is high risk, the engine defaults to diorama” was drawn from a README, not the artifact, and was wrong.
- **Proposed abstraction over photorealism** on the strength of that error, producing a dark, grainy “latent space” theme that resembled none of the references.